ENGN4627: Robotics

Research Report

A review of the paper
AMI: Adaptive Motion Imitation algorithm based on
Deep Reinforcement Learning

Submitted for the fulfillment of 30% of the coursework requirements under special consideration

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A note to the reader

1. The reader must understand this report in the context of the Australian roadmap developed by the Robotics Australia Group [1] (previously Australian Center for Robotic Vision). The paper of interest is viewed with respect to the current crisis need for imminent reforms while integrating socio-economic capacity-building measures within the legislative frameworks and policies.
2. No font or style requirements have been specified; hence I have taken the liberty to change the spacing to maintain the page limit while not compromising on the overall readability of this report.
3. Throughout this report, I have addressed where need be to oneself as "we" though this work has been done individually.
4. Please note that this document has been developed in the LaTeXenvironment, and the bibliography has been composed in the IEEE Transaction style using the BibTeX functionality, which means that the cited representation followed is the author's or the publishing copyright holder's representation of the material cited.

1. Executive Summary

This report aims to highlight the relevance of the paper "AMI: Adaptive Motion Imitation algorithm based on Deep Reinforcement Learning", published in the conference proceedings of the IEEE International Conference on Robotics and Automation, 2022 [2]. This report summarises, critiques, and builds on the social impacts of the results achieved by the Louisville Automation & Robotics Research Institute, University of Louisville, USA, supported by the US National Science Foundation. This work has been undertaken by Dr Nazita Taghavi (first author) and Dr Moath Alqatamin, with the practical assistance of Dr Sumit Das under the Social Robotics group headed by Prof. Dan Popa, Senior Member of the IEEE.

This paper introduces the AMI algorithm, a supervised motion imitation algorithm used by patients for physiotherapeutic treatment for those who have either lost appropriate motion speed or functionality owing to physical and mental disabilities, including but not limited to patients within the spectrum of autistic diseases [3]. The AMI learns the sequence by understanding shape and speed factors, tailoring it to individuals per their limitations or needs. The AMI algorithm owes its success to the Deep Deterministic Policy Gradient (DDPG) algorithm published in the seminal paper (over 10,000 citations) by Dr Lillicrap and others, Google Research. The paper builds on continuous control methods using an actor-critic architecture of policy development [4] instead of model-based methods in Deep Reinforcement Learning [5]. The upper-arm human motion imitation algorithm requires preprocessing of observed motion sequences followed by Z-normalisation [6] of the dataset. The data further requires a Fourier series embedding [7] to create a reward policy based on a Euclidean metric to utilise the DDPG algorithm. The algorithm trains the robotic system to imitate the learned sequence using deterministic policy gradients for episodic reward maximisation. This algorithm also uses the Dynamic Time Warping (DTW) cost as a measure of the accuracy of the learned and emulated sequences [8] [9] to make appropriate decisions in the instruction period. Thus, the AMI algorithm makes the robotic system robust to noise by synthesising reward-reinforced motion sequences under supervision. It uses appropriate verification and validation [10] in its implementation in two phases. The offline validation using motion capture on the Kinect® sensor is followed by the online implementation for the playback of the learned sequences. The algorithm uses a Python socket to emulate simple motions using the Robotic Operating System (ROS) for the Baxter and Zeno robots. As a necessary part of the unit testing, the imitation is emulated under safe robotic conditions by adapting the frequency and motion ranges for the patient who chooses to practice the exercise. It does this by learning from the quality of motion executed with respect to a default motion.

Finally, we explore the social and commercial impacts of the future of assistive robotics for a diverse range of issues in contemporary robotics communities. We focus on the possibilities this algorithm and its successors might bring from an Australian systems engineering and social perspective.

2. Motivation

This paper is replete with systemic and social motivation to influence the development of supervised learning in social robotics using model-free reinforcement learning methods for assistive physiotherapy. The field of work is at the intersection of imitation learning and assistive robotics. From a computer science perspective, imitation constrains the search space for learning leading to rapid outcomes in situations where designing the optimal motion for assistance is computationally hard. Human-Robot Interaction (HRI) principles have been here for some time, while Reinforcement Learning (RL) techniques are pretty recent. RL essentially engages in personalising interaction using optimal policy by tuning the actions for state interactions [11]. There is special attention to the development of such systems for patients with ASDs[1], where tailoring response and adapting to change is critical to the HRI, rehabilitation, correction or assistance [13]. Over the years, the highest and most direct impacts of imitation learning and assistive robotics have been in the area of autism research. These historical developments include the usage of neural networks for tracking joint angles for reaching a destination [14], the usage of predictive algorithms to match behaviour patterns [15], and using motion prediction by using visual descriptors [16]. Interested readers might find the detailed historical development review given in [17] useful. Despite such efforts, there are overarching bottlenecks, including socio-political and legal issues, which deter and stagnate the development of these systems because the lifecycle cost of such projects is either high or is not the priority of industrial decision-making bodies [1]. However, with a renewed understanding of HRI, which replaces costly, high-risk and repetitive tasks, large amounts of funding are flowing into research and development in this work area.

It has been observed that humans are often tired of training individuals with particular mirror neuron deficiencies or damages [18]. Therefore, social robotics is used to complement their exercise routines as necessary because they can learn sequences of motion and teach it while adapting to the difficulties faced by the patient [19]. In other words, these robots can treat partial accessibility over the joint movement space using imitation and adaptive sequence learning. It is followed by assisted teaching in

[1] Autism spectrum disorder (ASD) is a neuro-developmental disorder that causes significant impairment in her three main areas of function: communication, social interaction, and restricted, repetitive behaviours [12].

the instruction mode while being aware of the patient's shortcomings within the accessible joint space of the upper body [20]. In this context, time series analysis is a perfect method to learn the sequence of motion performed over time by analysing the similarity, nature, shape and frequency of the sequences learned over time. Rose et al. use this method to develop an end-to-end deep learning network that provides a larger input space for the actor to learn over the embedding provided for the available mappings. It uses the torque values at various pivots in the joint system leading to larger accuracy in the imitation learning process [21]. The DTW algorithm can thus be employed to learn a measure of accuracy and can be used as feedback to make decisions over time [9].

The AMI algorithm discussed in this paper borders on multiple works—the central motivation is a derivative of cyclical attractors theory for $\mathcal{S}^3$, $\mathcal{SO}(3)$ and $\mathbb{R}$ dynamic movement primitives (DMP) which analyse the coupled non-linearity in the movement of complex systems including joints [22]. This contribution is accredited to a long line of work undertaken by Ijspreet and others. They developed motion planning and trajectory modification using movement plans based on coupled nonlinear ODEs with limited and unknown attractor dynamics [23]. Structurally this field of work analysed the complexity from a biophysical perspective to imitate and learn the attractor dynamics themselves for robust and stable motion planning [24]. The motivation for exploiting the motion's cyclical nature to learn the default motion's shape and speed factors is inspired by the DMP approach in [25]. Tong et al. created an imitation learning benchmark by synthesising visual motion into the learning process of a small torque error joint movement of a 6-DOF robot.

With the advent of Deep learning, the robotics community has shifted gears to use neural networks and mainly reinforcement learning to analyse and recreate complex motion with multi-body constraints [26]. Deterministic Policy gradient methods [27] were limited to finitely discretised spaces making policy learning comparatively easy. This approach does not help take action in continuous spaces, as defining the accessibility for multi-linear systems is difficult. Discretisation methods in large networks suffer immensely from the curse of dimensionality [26], guaranteeing no fast convergence using iterative gradient descent methods [4]. This problem plagues almost all model-based learning methods where the reward shaping depends on the events in the action space and, therefore, a compact policy metric [26].

Model-free reinforcement learning methods are geared towards generative and semi-supervised methods that can work without a given environment and do not necessarily depend on the transitioning Markov state update. The approach is unlike Partially Observable Markov Decision Processes (POMDPs), where we find a decision process in which it is assumed that an MDP determines the system dynamics. However, the actor cannot access the underlying state and instead must build on the belief for a reward (or observation) in episodic batches. Only with the recent Deep deterministic policy gradient algorithm have we been able to generate continuous control policies in Euclidean spaces [4]. Three sources of inspiration for this paper are using RL techniques to create adaptive prostheses, and exoskeletons for assistive movement facilitation [28]. Secondly, a mirror therapy method using a master-slave interaction form for HRI was developed in [29], which inspired the two modes of implementation used in this paper. This paper derives its final contextual motivation from Di Febbo et al.'s [30] work. They use functional electric simulation [31] to stimulate injured muscles and produce an optimal controller to help patients extend their elbows to make the joint space accessible.

While motion planning is a separate field with its context in robotics and is infused with the use of RL [32], imitation learning finds its purpose in high-risk and often tailored requirements suitable for diverse use. One of the cheaper computational methods is understanding the inverse reward methods to find a sub-optimal action, given a state from a shaped reward, fast [33]. However, the promise of speed brings results while compromising on robustness and safety standards. This insight promotes supervised data preprocessing to create a robust set of data for sequence imitation to match a learnt sequence within a continuous control environment. Thus, we use the DDPG algorithm, which creates an actor-critic model for the target network to optimise the loss. It further updates the actor policy to evaluate the choice of action a given the states s for the policy network π for an iterative reward choice. To "critique" the policy optimality, the Bellman formulation uses a discount factor for an episode [4]. This factor weighs the feedback evaluation for the observer network $Q(s, a)$. The choice of reward using the Fourier-matched model provides appropriate feedback for updating the state response to the action, given the policy gradient in that episode [34]. Allowing noise within these minibatch updates leads to robust policies for the target networks. This means that given the shape and speed of the sequence transition, the algorithm can generate robust sequences for the motion estimation (immaterial of the choice of initialisation) [4].

3. Background

Before discussing the contributions, results and the necessary critique of this paper, it is essential to discuss a few concepts that have been used to develop the algorithm as a measure to understand the motion and root of the proposed algorithm. In this section, we shall discuss the formulation of the adaptive motion imitation algorithm from the DDPG algorithm, the Fourier

coefficients learning to form the shape and speed coefficients using robotic vision and the DTW algorithm, which is used to find the accuracy of learned sequences in the adaptive mode.

3.1 Fourier approximation of the Joint space

A Fourier series is the expansion of a periodic function f(x) for the infinite sum of sines and cosines. It utilises the orthogonal relationship between the sine and cosine functions. The computation of the Fourier series is very useful for decomposing any periodic function into a simple set of harmonic terms. One can solve and recombine them to get a solution to the original problem or approximation to the required or practical precision. In general, Fourier series are assumed to converge almost everywhere (except at discrete discontinuities). This simplification arises where functions encountered in engineering behave better than mathematicians can offer as counterexamples to this assumption. In particular, if the function $f(x)$ is continuous and the derivative of $f(x)$ (which may not exist anywhere) is integrable, then the Fourier series of f uniformly converges to $f(x)$. We exploit this property to find the coefficients given a particular choice of accuracy, the joint space to be accessed by the multi-body mechanical system to a Z value normalised dataset, which is scale-invariant for the motion under study.

In the steady state, the robot implementing the process can therefore mimic the motion to a high degree of accuracy. This process is possible theoretically for any motion; however, for cyclical and periodic motions, this motion form is much simpler because it is easier to develop a singular model that, upon parameter variation, can exhibit a range of motions mainly used in physiotherapy exercises. Thus, when the desired trajectory for every subpart of the observed motion has only a finite duration, it can be approximated by a Fourier series with constant frequency magnitudes. For each subsystem of the robot, a learning controller is designed to individually control each harmonic component of the actual output, although it is cross-related to other components in nonlinear systems.

For each robot subsystem, a learning controller is designed to independently control each harmonic component of the actual output, but in nonlinear systems, it is interrelated with the other components [7]. Instead of finding the exact dynamics, it is much simpler to approximate the resulting motion. Using the Gibbs theorem, we may analyse every function as a Fourier approximation in terms of the harmonics making the modelling much simpler. Hence, the authors define only two system parameters—the shape factor α_{ij} and the speed factor w_{ij}, which are embedded into each motion indexed as j for every motion sequence i. This approach leads to the joint space angle sequence F_m, given using the harmonic coefficients $a_{i,j}$ as:

$$F_{m,j}(i) = a_{i,0} + \alpha_{i,j} \sum_{i,t=1}^{j} \left(a_{it} cos(w_{i,j} i) + b_{it} sin(w_{i,j} i) \right). \tag{1}$$

3.2 Deep Deterministic Policy Gradient (DDPG) algorithm

The DDPG algorithm is a Deep Reinforcement Learning (DRL) method that simultaneously learns the Q-function and policy. Train the Q function using out-of-policy data and Bellman's equation describing the optimal action-value function, $Q^*(s,a)$ [4]. It's given by

$$Q^*(s,a) = \underset{s' \sim P}{\mathrm{E}} \left[r(s,a) + \gamma \max_{a'} Q^* \left(s', a' \right) \right], \tag{2}$$

where $s' \sim P$ is shorthand for saying that the next state, s', is sampled by the environment from a distribution $P(\cdot \mid s,a)$ and γ is the discount factor which determines the weight on the learnt Q-function[2]. Knowing the optimal action-value function $Q^*(s,a)$, the network can find the optimal action $a^*(s)$ in any state by solving the following equation:

$$a^*(s) = \arg\max_a Q^*(s,a).$$

The DDPG algorithm uses nested training for the approximator $Q^*(s,a)$ with the training of the approximator $a^*(s)$. This method is particularly well suited for environments with a continuous action space. If there are a finite number of discrete actions, the maximum value is not an issue because the Q values for each action can be calculated separately and compared directly. (This also immediately gives the action that maximises the Q-value.) However, if the action space is continuous, the space cannot be exhaustively evaluated and solving the optimisation problem is very difficult [26]. Using standard optimisation algorithms, computing $\max_a Q^*(s,a)$ would be a very expensive subroutine. Also, this becomes infeasible as it needs to be run every time the agent learns something new in the environment. Since the action space is continuous, the function $Q^*(s,a)$ is considered differentiable for the action argument [4]. This process allows us to build an efficient gradient-based policy learning rule $\mu(s)$

[2] This simplified formulation has been adopted from https://spinningup.openai.com/en/latest/algorithms/ddpg.html.

that exploits this fact. Instead of running an expensive optimisation subroutine every time we compute $\max_a Q(s, a)$, it can be cheaply approximated with

$$\max_a Q(s, a) \approx Q(s, \mu(s)).$$

DDPG trains deterministic policies in an out-of-policy way. Policies are deterministic, so when an agent examines a policy, it may not initially attempt a wide range of actions to find helpful learning cues. Addition of noise to actions during training to better explore DDPG guidelines. The authors of his original DDPG paper recommended time-correlated OU noise [4], but recent results suggest that zero-mean uncorrelated Gaussian noise is a good enough substitute.

3.3 DTW algorithm

Dynamic time warping algorithms measure temporal similarity between time series or continuous data sequences [35]. DTW is generally a method of computing the best fit between two specific sequences (such as time series) using certain specifications. "The algorithm must match each index in the first sequence with one or more indices in the other sequence and vice versa. The algorithm must match the first index of the first sequence with the first index of the other sequence. The algorithm must match the first sequence's last index with the other sequence's last index. The mapping from the indices of the first sequence to the indices of the other sequences should increase monotonically and vice versa"[3].

The DTW cost is computed as the sum of the absolute differences of each pair of matched indices, and the best has the lowest cost. The sequences are warped non-linearly in time to determine a measure of their similarity independent of certain nonlinear variations in the time dimension. This sequence alignment method is often used in time series classification. A DTW measures quantities like the distance between two given sequences, but it doesn't guarantee that the triangle inequality holds and that the cost of the DTW is a measure rather than a metric [35] [4]. In addition to a similarity measure between two sequences, a so-called "warping path" is generated. The two signals can be aligned in time by distorting them along this path. A signal with an original set of points X(original), Y(original) is transformed to X(warp), Y(warp). The cost is then calculated between them as a measure of the aligned difference in waveform or sequence pattern. The DTW algorithm provides a basis to compare sequential data based on their behaviour dynamics rather than the point-to-point difference with respect to the time parameter.

4. Discussion

This paper develops an adaptive motion imitation algorithm (AMI) based on the DDPG algorithm. This synthesised algorithm is implemented on the Baxter and Zeno robots to assist in physiotherapy motions while learning from specific demonstrations and synthesising that to the patient's capabilities. This paper's original contribution is finding a reward metric that compares the Fourier series coefficient learning for emulating the speed and shape to best match the quality and speed of the subject using the DDMP algorithm using visual inputs to learn the cyclic joint motions. Motion quality is assessed using the online DTW algorithm, which stems from the previous work of the same authors using sequence buffering techniques [9].

From the overall context of social robotics, this paper's original contribution develops a measure as feedback to understand the motion quality with respect to the speed and shape of the movement. This development serves as a guide for future research to extend imitation to various parts of the body. Patients with ASD or other disabilities would benefit from the physiotherapeutic methods in the instruction mode of the algorithm's execution by improving the motion quality, personalised to the users' requirements. While the execution and the technicality of this paper are not a milestone development, it does answer a few open problems in assistive robotics to build optimality criterion as a benchmark for rehabilitation via HRI [34]. As discussed in the motivation, reinforcement learning provides a robust method to analyse these imitation movements. The robot can develop profiles using the continuous updates of the reward evolution rather than relying on computationally expensive feature descriptors or behaviours to model its motion. From the computational perspective, this paper introduces a novel method that bounds the search space by building critic-reinforced architecture for much more accurate results (pending theoretical guarantees).

Finally, this paper provides a hopeful closure to whether inferring intent is necessary to explain particular behaviours [36]. Model-free RL methods do not require goal or intent establishment for developing a method for imitation because instead of relying on goals [26], it relies on penalty-reinforced behaviour approximation in dynamical frameworks. Thus, the necessity for imitation based on intention (an open question in this area) finds some resolution (albeit not absolute) because the community now has the answers to when and how to imitate human responses or interactions without overtly relying on targeted activation methods [17].

[3] This insight has been borrowed from the Wikipedia page: `https://en.wikipedia.org/wiki/Dynamic_time_warping`

[4] However, that assumption is often bridged to accelerate the algorithm.

4.1 The AMI algorithm

The Adaptive Motion Imitation (AMI) algorithm [2] is built on the DDMP algorithm [4] as discussed previously. The algorithm requires preprocessed data to fit the Fourier series modelling in the joint space. If the kinematics differ for humans and robots, the motion data captured is transformed into a joint space corresponding to the robot's inverse kinematics to create a scale-invariant embedding of the human arm to the robot's joint space. Thus, the extracted joint angle-based actuation is mapped as a function of the robot's inverse kinematics. This preprocessing allows for creating a measured sequence for a given time-step $\phi_j(i)$ for every joint indexed as j.

The algorithm creates a DDPG actor-critic network and another dedicated neural network for the buffering process. It initialises the weights for these networks. It also creates a target network for the interpolation sequence of the DDPG algorithm. The algorithm defines a shape and speed factor for every iteration along with random noise to make the modelling robust. The DDPG algorithm trains the robotic system for the target motion trajectory in joint space. The architecture has two distinct neural networks, one representing the policy network $a^* = \pi(s)$, and another the actor-critic architecture $Q(s, a^*)$. A separate reply buffer stores the transition rewards for the next episode. The two other target networks defined for the DDPG algorithm update the strategy corresponding to the actors and approximate the state's action-value function, given the critic's response to the policy in that episode. The number of time steps is kept equal to the length of the sequence recordings, set up on a maximum of three cycles from the data set. A Euclidean distance-based 1-norm reward $R_{ij} = -|\phi_j(i) - F_{mj}(i)|$ is calculated for each joint in every episode, such that the policy is updated by the episodic rewards $R_{tj} = \sum_{i=0}^{I_t} R_{ij}$ for every time-step I_t. The critic, actor and target networks are updated accordingly using a sample of a mini-batch transition and the Bellman Q-learning function (2). The policy update is loss minimised for every episode for a given choice of the discounting parameter. The actor policy is updated using the sampled policy gradient given the policy choice, input states, and a Polyak interpolation factor (based on Polyak averaging) [27]. This is used to update the target networks once per actor-critic network update. The AMI algorithm calls back the values obtained for the actor and critic for the next episode. This process goes on iteratively till the DDPG algorithm converges for every joint j of the study.

4.2 Results

In the adaptive mode, the robotic system observes the motion of the subject and creates an action tuple consisting of the shape and speed factors $a_{tj} = \{\alpha_{ik}, w_{ij}\}$ to embed the Fourier function (1) for the entire episode of the selected dataset I_t. When the subject performs a motion with constant speed and shape throughout the action, the robot records approximately the same action values for every time-step datapoint in I_t. After the robot learns the subject's motion, it calculates an average action model for the sequence and adapts the speed and motion to that of the subject. The authors show that after three cycles of default motion study, the shape and speed factors converged to the values 1.1 and 0.31, respectively, with maximisation of the policy reward, after 500 episodes. When another subject performs the same motion at slower speeds, we can see that the robot learns that sequence and adapts the shape and speed factors appropriately. It is observed that the sequence data needs to be sent to the robot joint at 22 ms to match the human speed. After the training period, the quality of the learned sequence is tested using the DTW algorithm for the eleven subjects who acted at slower speeds. Results revealed that the DTW cost for the default observed motion and learned sequence is significantly smaller than if the same motion is learnt fast.

For patients who want to practice a motion for physiotherapeutic purposes, the instruction mode generates easy motions with respect to the observed default motion. As the patient's motion quality improves (characterised by the DTW cost percentage decrease), the robot switches to a more complex motion for the patient, gradually increasing their capabilities to access the joint space freely. To test the instructor mode, the authors use a 15-pound weight for subjects to mimic the incapabilities of the patients while performing the same motion as in the adaptive mode. It is observed that the rewards maximise and the action tuple values converge to certain constants after several episodes of the algorithm's episodic iterations. However, the subject(s) were empirically unable to control the constant speed and shape factors in their motion. This result shows that despite the non-homogeneity in motion, the algorithm can capture the average action values and find convergence to a near approximate in the offline validation. The result shows that after repeated experiments on seven subjects, the action tuple values converge to a constant (approximately the average of the speed and shape factors for the entire study). Until the subject's motion quality improves, the robot selects a range of motion with lower speeds and ranges, which are easier to practice.

To reaffirm the results obtained in the offline implementation, the author's playback the learned motion sequences on the Baxter robot in the online mode of learning sequences. The authors intend for the robot to learn a default motion sequence (in this case, a handwave) and mimic it depending on the algorithm's reward threshold defined for that task. Note that the robot learns the sequence and executes it through a socket connecting the online AMI algorithm to the ROS of the Baxter Robot to realise the action response in real-time. This experiment heavily relies on the ROS's query execution in the standby mode to simultaneously

passes joint commands while it polls for inputs for the joint angles.

The execution uses a bounded safety configuration with respect to the inverse kinematics of the robot and the lab environment. As the subject starts the motion, the algorithm picks up the motion, and the DDPG network calculates the actor-critic evaluation to find the online sum of the episodic rewards contiguously. Once the reward threshold is reached (heuristically selected), the robot mimics the motion in real time. Once this reward is consistently higher than the threshold, the joint command updates the episodic reward to the highest observed and continues the motion on the joint commands of the last best reward. The reward shaping improves progressively, the motion imitation online, and the average action pair tuple evolves to update the joint commands if an episodic reward is observed, which is higher than the current best.

5. Critique

This paper suffers from interspersed ambiguities in experiments. While we shall discuss them in detail, it is essential to address the particular issue of academic integrity. The use of the Fourier series to model the motion are not original to this paper; merely using it in this context is new. However, the technique has yet to be referenced in previous work. It was introduced by Tang et al. [7] much earlier as an alternative to DMPs.

The paper needs to justify parameter evaluation for the DDPG algorithm integration. It needs to provide the relevant cross-studies to compare the choice of parameters while formulating the algorithm in the adaptive and instructor modes. The discount and interpolation factors play critical roles in the success of the DDPG algorithm's results for episodic rewards, which act as feedback for the critic network; therefore, we expect a detailed cross-study to make this approach more explainable even if it is from a heuristic perspective. Using a 15-pound weight to emulate difficulty reproducing the quality of motion requires further justification.

The paper expects us to take at face value the central insight that it is easier for the algorithm to adapt to slower motions, which is intuitive but does not clarify the hardware limitations and whether they could potentially affect the results obtained. It needs to specify the subject's capabilities in detail. These ambiguities make the study empirical at best. The algorithm can adapt to the generated sequences, but we need a detailed study of the Fourier representation of the original motion to verify the error rates for the original motion with respect to the Z-normalised data of the learned sequence in the scale-invariant joint space. The studies which should supplement the results have neither been appended nor referenced to any external sources making it difficult to verify or reproduce. Thus, the authors must include essential details in the online implementation and the playback of learned motions.

The authors do not specify the safety limitations for the bounds in terms of joint space and torque applied, nor does it define the criterion for "properly learning the trajectory". Much of the work in the online playback for the episodic choice of reward is eyeballed rather than quantitatively analysed with respect to the episodic rewards reached by the algorithm for every learned sequence playback. This issue plagues the entire study with online verification for the elbow joint for the Baxter robot. It is essential to realise that changes not visible to the human eye between the episodic rewards of -35 to -40 improve the robot's motion.

Finally, there is no study of the time or space complexity of the algorithm's execution in various modes, and neither is it trivial (for example, a gradient descent algorithm has known step-size complexity, which is cross-studied and established). This void instigates questions on the deployment readiness of this algorithm.

6. Social Impact

A critical social impact of this research line is providing physical and cognitive support to people with diverse needs. The current research has explored various aspects of HRI and its applications in various fields, such as education and patient care. The integration of robots into the healthcare system is known as robot therapy, which is used as a substitute for animal-assisted therapy [15]. However, to assess the acceptability and effectiveness of social robots and use them as viable replacements and understand their impact widely in diverse scenarios of human interactions, these robots need to be deployed in the "wild" field for an extended period. Such engagement typically includes collaborations with fields such as medicine, social psychology, clinical therapy, industrial design, public health, marketing and education.

Therefore, this research area addresses the investigation of social robotics using novel algorithms and computer modelling that is being evaluated by intended users/consumers, patients, or individuals with special needs. This makes Social Assistive Robotics (SARs) a significant initiative in the process of robotics integration into society. SAR deployments require massive collaborative efforts and hence demand interdisciplinary research. While we proceed cautiously in this area, we find that SARs have the

unique opportunity to integrate into the daily lives of individuals with ASD, mechanical disabilities, and internal medical issues in certain situations. For example, stroke-recovering patients or patients with reduced neural motor transduction may require an exoskeleton or implants to control their muscles to correct their gait. This is where modern machine learning systems are beneficial. RL methods use penalty based rewarding to appropriate the correct response, making it much more feasible to train these systems without finding the computationally expensive dynamics of the motion [21].

While the dimensionality issue does not plague model-free RL frameworks and longer [4], concerns involving explainability in deep learning frameworks make engineers deploying such systems hesitate to make solid guarantees on the safety, and sanitary limitations [37]. This crisis spearheads much of the work on the explainability of AI systems, which, to crack open the black boxes, engineers and researchers will have to proceed cautiously [38]. Empirical studies suggest that this inaccessibility of AI systems makes them unreliable for use in healthcare systems. Hence parallel research on the development of fundamental research on various open systems perspectives is crucial.

One concern for social roboticists is the component of reciprocity in HRI. Reciprocity, while a concern for significant scientists to develop in SARs, the components of HRI, can be enhanced by using reciprocity as a reward in designing reinforcement learning systems. However, the multi-dimensional aspects of what humans might consider as natural rewarding behaviour are challenging to emulate until we may create augmenting reward systems. These systems would take into account bias, reasoning and thought evolution. The process of mimicking this from human-human interaction is often criticised but might be effective in understanding emotional engagement in social robotics. The only risk of this approach would be not to make the robot's policy evolve based on these external rewards via reciprocation because it threatens to end the willingness to execute tasks that would not provide similar reciprocation. Designing for reciprocity in social robotics should, thus, be in tandem with reciprocation as a tool to enhance performance in complex tasks rather than a bidirectional consequential policy [39].

This paper paves the way to develop new algorithms and robot therapists for physiotherapy instruction and assistance in SARs. The AMI algorithm uses DRL to adapt to the patient's capabilities and adjusts motor skills to match the challenges faced in the upper body limbs for the exercises. This approach allows for low-fidelity yet safe engagement for patients with limited or damaged motor skills, especially for patients with autism, making it a safe choice. Because of the unique approach to shaping feedback rewards by comparing the Fourier series model to the measured sequence for the joint, followed by the DTW algorithm-based testing, this contribution sits within the priority outlined in the Australian roadmap, 2022 [1]. We refer to the priority stance "Increase research and development and improve commercialisation of locally developed intellectual property in robotics and robotics-related technologies", where the following ethical principles are central to our recommendation:

- Human, social and environmental wellbeing: Throughout their lifecycle, AI systems should benefit individuals, society and the environment.
- Human-centred values: Throughout their lifecycle, AI systems should respect human rights, diversity, and the autonomy of individuals.

We believe that the work undertaken in this paper shall enhance human lifestyle, health and accessibility while maintaining the sustainability of a diverse set of health sector enterprises where robots reduce the workload on healthcare professionals by tackling repetitive tasks. "The main benefit of these technologies is their ability to act as a force multiplier in areas traditionally under-resourced" [1]. As the authors suggest, future research will refine the AMI algorithm with time in its implementation. These improvements, as suggested, will account for high-fidelity movements and robust and high-performance controllers for adapting faster (lowering training time) to the sequence measurements.

From a social and commercial viewpoint, this algorithm and its potential successors shall evolve into hybrid methods to build reinforcement learning methods that can adapt to a more diverse range of functionalities that commercially impact policy-making in healthcare sectors and therefore need more interdisciplinary participation. Along with higher participation, the economics of SAR usage shall become increasingly feasible and will potentially complement traditional methods of treatment as it has been found that ASD patients often perform better with robotic assistance than human instruction [40]. The social impact of this would be the wearing out of the stigma associated with learning robots in households and hospitals.

In conclusion, this line of research by the University of Louisville serves cross-sectional and intersection of different groups going through the treatment process. From the Australian context, we must become the developer and producers of SARs instead of buying or technology transfers because of the crisis in healthcare sectors from the sheer lack in the number of healthcare professionals. This plan is feasible because of the large pool of talent in Mechatronics and AI in Australia who are industry-ready. Rapid modernisation to integrate SARs as a treatment convention will lower the pressure on the small group of professionals to make healthcare more equitable. However, we need rapid policy transformation in terms of legislative and financial frameworks to catch up with the rest of the world in this area.

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